

Counting What the Classifier Sees: Measurement Validity in the Study of AI, Digital Platforms and Extremism

Research proposal for the PhD Research Fellow position in Information Systems (Jobbno 307398), University of Agder · Asad Ikram · September 2026

1. SUMMARY

Every published claim about how much extremist content circulates on a platform, and about whether an algorithm amplifies it, is a ratio. The numerator is produced by a classifier, human or machine, and the classifiers in this field do not agree with each other, do not generalise across datasets, and change under our feet when a vendor updates a model. This project treats that as the research problem rather than a footnote. It asks how the sociotechnical arrangement of platforms, AI systems and governance institutions produces the numbers that policy relies on, and it builds and tests the measurement discipline needed to make those numbers comparable across platforms, languages and time. The work is anchored in the Norwegian setting: the national centre against violent extremism (NIVE) has been mandated to develop a common risk assessment tool at exactly the moment the best available systematic review finds that no existing tool of that kind has demonstrated predictive validity. The thesis combines an Information Systems reading of platform affordances and institutional logics with psychometric methods for machine annotation, and delivers a validity protocol that a centre like NIVE can use.

2. BACKGROUND AND PROBLEM

The Department's call asks how digital platforms feature the construction and circulation of radical content, how algorithms amplify or suppress it, and how these dynamics can be regulated. Information Systems has only recently claimed this ground. Risius, Blasiak, Wibisono and colleagues (2024) reviewed 222 articles on the digital augmentation of extremism and argued that IS is well placed to study extremism as a sociotechnical phenomenon, in which technical features, user practices and institutional responses shape one another. I take that agenda as my starting point, and I take one of its blind spots as my problem.

The first-order question, whether recommender systems radicalise, has been asked across several generations of platform design, and the answer is now contingent and well documented. Hosseinmardi and colleagues (2024) estimate causally that after the 2019 changes YouTube's recommender is, if anything, a moderating force on solitary extremist viewing. Chen and colleagues (2023) show that exposure to alternative and extremist channels concentrates among users who already hold resentful attitudes and arrives through subscriptions and external links rather than recommendations. Horta Ribeiro, Veselovsky and West (2023) explain mechanically why sock-puppet audits and user-trace studies contradict each other. Haroon and colleagues (2023) find that problematic recommendations are rare as a share of all recommendations, around 2.5 percent, yet reach more than a third of their audit accounts. These are careful studies and I do not propose to re-litigate them.

What I propose to study is what all of them share. Each estimate of prevalence, exposure or amplification depends on a classifier that decides which items count as extremist, radical, problematic or harmful. Yin and Zubiaga (2021) showed that the cross-dataset generalisation of abusive-language detectors has been severely overestimated: models that look accurate on

their own test set lose much of that accuracy on a neighbouring one. Riabi and colleagues (2025) showed that annotator disagreement on radical content is systematic and patterned by who the annotators are, not random noise to be averaged away. Vidgen and Derczynski (2020) documented how training-data choices propagate into every downstream figure. None of this is unknown to the authors above, who report validation statistics conscientiously. The gap is that the field has no accepted way to carry classifier disagreement and instrument change through to the final estimate, so a prevalence figure is published with the classifier's error treated as fixed, small and platform-neutral, when the evidence says it is none of those things.

I call this the denominator problem, because the same instrument that decides the numerator also silently decides what population of content was eligible to be counted, and because the failure is invisible in the published number. My own work gives me reason to think the effect is large. In a methods paper written with Dr Ella Haig (Portsmouth), we treat each social platform as a survey mode and apply measurement theory to machine-labelled text. In a simulation with five hundred replications and known ground truth, an index that pools platform-level scores at face value reports the wrong sign of the aggregate on roughly a quarter of days, purely from measurement differences between platforms. The paper is about economic sentiment, but the machinery is agnostic to the construct, and extremism is the construct where the cost of a wrong sign is highest.

The Norwegian setting makes this timely rather than merely interesting. The Extremism Commission (NOU 2024:3) and the subsequent white paper (Meld. St. 13, 2024-2025) set out a programme of measures, and the new national centre, NIVE, is charged among other things with developing a common risk assessment tool for the police security service, the police and frontline services. In December 2025 a Campbell systematic review of six violent-extremism risk assessment tools (Brouillette-Alarie, Hassan and colleagues) found that no true predictive validity had been established for any of them, that the designs were retrospective, and that field inter-rater agreement was markedly lower than in research settings. A national centre is about to build an instrument in a family of instruments that has not yet been validated. The measurement question in my proposal and the practical question in NIVE's mandate are the same question at two levels of the system, and the Department's expectation that the fellow collaborates with NIVE is the reason this project belongs here.

3. THEORETICAL PERSPECTIVE

I approach the problem as sociotechnical IS, with the measurement machinery as method rather than thesis. Three perspectives do the work.

Affordances and institutional logics. Thapa, Thapa and Sæbø (Information Systems Journal, doi 10.1111/isj.70018) show how information-systems initiatives intended to reduce inequality can aggravate it when the affordances a system offers meet institutional logics that pull in a different direction. I read the circulation of radical content the same way. A platform offers affordances for construction and circulation: recommendation, subscription, search, cross-posting, external linking, ephemerality, encryption. Those affordances are actualised differently by users seeking radical material, by moderators operating under a trust-and-safety logic, by regulators operating under a public-safety logic, and by researchers operating under a scientific logic that needs stable constructs. A classifier is itself an affordance-bearing artefact: it makes some content countable and other content invisible, and the counts it produces then travel into transparency reports, risk assessments and policy. The framework

lets me ask not only how much circulates but which logic decided what circulation means, and where those logics conflict.

Measurement theory. Classical test theory, generalizability theory (Brennan, 2001) and measurement invariance testing (Putnick and Bornstein, 2016) were built for exactly this situation: an unobserved construct, several imperfect instruments, and a need to know whether the instruments measure the same thing across groups. Platforms, languages, annotator populations and model versions are groups. Invariance tests ask whether a score of X on platform A means what it means on platform B. Generalizability studies decompose disagreement into named sources, so that annotator standpoint, language and platform each get a variance component rather than being folded into "error". Score linking and equating let a series survive an instrument change. These are standard tools in educational measurement and survey methodology that have barely been applied to machine annotation of platform content.

Design science. Following Hevner, March, Park and Ram (2004), the final contribution is an artefact: a validity protocol and reporting standard for extremism monitoring, built with a practitioner organisation and evaluated in use. IS has a tradition of contributing artefacts that carry theory into practice, and a national centre building a risk tool is an unusually direct route for that.

4. RESEARCH QUESTIONS

RQ1 (Equivalence). Do the classifiers used to identify extremist and radical content, whether trained models, large language models or platform-reported labels, measure the same construct across platforms, across English and Norwegian, and across annotator standpoints? Where they do not, what is the size and direction of the difference?

RQ2 (Amplification). When classifier disagreement, cross-dataset error and model drift are carried through to the estimate, do published conclusions about algorithmic amplification and suppression of radical content survive, and which platform affordances actually carry circulation?

RQ3 (Governance). How do the institutional logics of platforms, regulators and NIVE shape what counts as extremist content and what counts as valid evidence of it, and where do those logics conflict with the measurement discipline that stable monitoring requires?

RQ4 (Artefact). What would a validity protocol for extremism monitoring and risk assessment look like, and does it improve the comparability and defensibility of the numbers when adopted by a practitioner organisation?

5. STUDIES AND METHODS

Study 1. Equivalence: do the instruments agree, and where do they disagree? (RQ1)

Study 1 assembles a set of instruments and tests them against each other on a shared corpus. The instruments are of three kinds: open-weight classifiers trained on public extremism and abusive-language datasets, large language models prompted with codebooks, and platform-supplied labels where transparency reports or APIs expose them. The corpus is a stratified sample of public content drawn from three platforms with durable researcher access, in English and Norwegian, spanning content that the literature classifies as extremist,

borderline and benign. A subset is annotated by a panel recruited to vary in standpoint, following Riabi and colleagues' finding that disagreement is patterned by annotator background, with a codebook co-developed with NIVE practitioners so that the construct being measured is the one the Norwegian system actually uses.

The analysis fits configural, metric and scalar invariance models with platform and language as grouping variables, tests differential item functioning at the item level, and runs a generalizability study with annotator standpoint, language and instrument as facets. The Norwegian side uses the NB-BERT and NorBERT model families and the NorBench evaluation suite, together with existing Norwegian hate-speech datasets, so that the language comparison is between properly resourced instruments rather than between English and a translation. Outputs are a set of calibrated prevalence estimates whose intervals reflect instrument disagreement rather than sampling error alone, and a public table of which instrument pairs are exchangeable and which are not. I expect this to be the first fitted invariance analysis of extremism classifiers with platform and language as grouping variables.

Study 2. Amplification with the error carried through (RQ2)

Study 2 re-estimates amplification and exposure on two platforms using both audit and trace designs from the literature, and propagates the measurement model from Study 1 into the estimate. The audit arm follows the sock-puppet logic of Haroon and colleagues, with accounts that differ in seeded ideology and consumption history. The trace arm uses public engagement and subscription signals in the style of Chen and colleagues, within the limits of the access described in section 6. The difference from prior work is not the design but the estimator: every classified item carries a stratified misclassification correction estimated in Study 1, so that an amplification ratio has an interval that includes what the classifier does not know.

Two further elements address instrument change over time. First, a pinned open-weight shadow annotator runs in parallel with any vendor model throughout the project, so that when a vendor updates, a bridging estimate is available from the overlap. Second, a version ladder and comparability ratios let the series be equated across changes, in the manner of survey mode transitions. I run this discipline commercially already, with weekly divergence alarms on production classifiers, and I have seen what happens without it: a series that moves because the instrument moved. The affordance analysis sits on top of the corrected estimates: circulation is attributed to recommendation, subscription, search, external linking and cross-posting, so that the question shifts from "does the algorithm amplify" to "which affordance carries what share of circulation once the instrument is accounted for".

Study 3. Governance and institutional logics (RQ3)

Study 3 is interpretive. It combines document analysis of platform transparency reports, Digital Services Act risk assessments, the Extremism Commission report and the white paper, with interviews with practitioners at NIVE, in the police and frontline services, and, where access permits, in platform trust-and-safety teams. The analytical lens is the affordance-based institutional-logics framework, applied to the specific question of how each actor defines extremist content, what evidence they treat as valid, and how numbers produced under one logic are consumed under another. A regulator citing a platform's prevalence figure, for instance, imports the platform's classifier and the platform's definition, often without knowing it. The study maps those imports and the conflicts they create, including the

case that matters most to NIVE: a common risk assessment tool that must serve the security service, the police and social workers, whose logics differ and whose inter-rater agreement in the field is known to be lower than in research.

I am candid that qualitative interviewing is not something I have done in a research setting. This study is designed to be co-supervised by someone for whom it is native, and its scale is deliberately modest: fifteen to twenty-five interviews, coded with a template developed from the theoretical framework. The document analysis I can carry alone.

Study 4. The artefact: a validity protocol for extremism monitoring (RQ4)

Study 4 turns Studies 1 to 3 into a design-science contribution. The artefact is a validity protocol and reporting standard: a specification of what a prevalence or risk figure must carry with it to be interpretable (instrument identity and version, codebook, invariance status across the groups it is applied to, misclassification matrix, drift monitoring), together with a lightweight implementation that a centre can run. The evaluation follows the design-science cycle: build with NIVE as the practitioner partner, evaluate in use against the centre's own monitoring needs and, if the timing allows, against the development of the common risk assessment tool, where the Campbell review's findings on retrospective designs and field inter-rater agreement give a concrete list of what the protocol must address. The measure of success is whether two analysts using the protocol on the same data produce comparable figures and can say why they differ. I have built the operational version of this for commercial classifiers; the research contribution is to specify it, ground it in measurement theory, and test it where the stakes are public safety rather than brand analytics.

6. DATA, ACCESS AND INFRASTRUCTURE

The project uses three platforms with durable researcher access, chosen with the supervisor in the first months. The candidates are YouTube through its Data API, Reddit through its API and public archives, and public Telegram channels through the public API, with TikTok's research access as an option if it is available to Norwegian institutions at the time. I do not assume vetted-researcher access under Article 40 of the Digital Services Act. The Act had not been incorporated into the EEA Agreement as of August 2026, and a proposal that depends on access that may not exist is not a feasible proposal. If Article 40 access becomes available during the project it extends Study 2; nothing depends on it.

Collection risk is the risk I am best placed to remove. I have spent seven years building production data collection: a fleet of sixty-eight collectors with an automated repair layer at M+C Saatchi Fluency, a team running over five hundred collectors across fifteen countries at Dubizzle Labs, and a public engineering reference at web-scraping-guide.com. For this project I will use official APIs where they exist and document every collection path in a provenance log, so that what was collected, when, under which terms and with which instrument is recoverable for any figure in the thesis. Norwegian-language resources are the NorBERT and NB-BERT families, NorBench, and the existing Norwegian hate-speech corpora; English resources are the public extremism and abusive-language datasets used in the literature cited above. All data are held on University of Agder infrastructure under the data-management plan agreed with Sikt.

7. ETHICS, SAFETY AND DATA PROTECTION

The ethics are designed in rather than appended. The legal basis for processing special-category data manifestly made public by the data subject is Article 9(2)(e) of the GDPR, combined with processing in the public interest, which Sikt's guidance treats as more appropriate than consent for research on public platform content. I will complete a data protection impact assessment under Article 35 before collection begins, and register the project with Sikt. NESH's test of reasonable expectation of publicity governs what is collected: public channels and public posts, no private groups, no circumvention of access controls. Because the call's population is young people, I treat minors explicitly: no individual-level profiling, aggregate analysis only, no attempt to identify users, and content from accounts that self-identify as minors excluded from any item-level release. No individual-level data leaves the university, and nothing in the project produces a list of people. The sock-puppet audit in Study 2 creates research accounts that never interact with real users. Interview participants in Study 3 give informed consent and are anonymised at the point of transcription.

Researcher safety matters in this field. Exposure to extremist content is scheduled, limited and shared, following the practice of established trust-and-safety research groups, and the annotation panel in Study 1 is briefed, supported and free to withdraw. Where NIVE data are involved, the project takes the centre's own classification rules as binding and publishes only what the centre clears.

8. PLANNED PROGRESSION

Period	Work	Deliverable
Months 1-6	Coursework front-loaded. Platform and instrument selection with supervisor. DPIA, Sikt registration, data-management plan. Codebook workshops with NIVE. Corpus collection begins.	Approved research plan (month 3). Ethics clearance. Corpus v1.
Months 6-18	Study 1: annotation panel, instrument runs, invariance and generalizability analyses. Study 3 document analysis in parallel.	Paper 1 submitted (month 18). Public instrument-exchangeability table.
Months 12-26	Study 2: audit and trace designs, shadow annotator running from month 12, corrected amplification estimates. Study 3 interviews.	Paper 2 submitted (month 26). Paper 3 drafted.
Months 24-32	Study 4: protocol built with NIVE, evaluated in use.	Paper 4 submitted (month 32). Protocol released.
Months 30-36	Thesis writing, kappa, defence.	Thesis submitted (month 34); defence by month 36.

The plan is written for the three-year track. On the four-year track with 25 percent career-promoting work the same milestones stretch by roughly a third, with teaching placed in months 7 to 30 where it disturbs the studies least. Decision gates at months 6, 12 and 24 carry pre-specified fallbacks: if a platform withdraws access, the design substitutes a second platform from the reserve list rather than shrinking the comparison; if NIVE's timetable for the risk tool slips beyond the project, Study 4 evaluates the protocol on the centre's monitoring activity instead. Papers 1 and 2 are aimed at venues with fast cycles in computational social science; Papers 3 and 4 at Information Systems journals, where the affordance and design-science contributions belong.

9. WHY I AM THE RIGHT PERSON TO DO THIS, AND WHAT I DO NOT BRING

I have built the instruments this proposal is about. At ArtemisAI, a company I founded, we labelled 3.39 million comments across a two-tier annotation system. The two tiers agree on 82.9 percent of sentiment labels, 74.4 percent of emotion labels and 0.0 percent of toxicity labels, while the toxicity model reports 96.7 percent average confidence. The zero was a mismatch of label taxonomies between tiers, not a failure of either model, and the dashboard's accuracy figure was inter-model agreement with no criterion behind it. Nobody built that dishonestly. It is what happens when instruments are adopted faster than the discipline to check them, and it is the same thing that happens when a prevalence figure travels from a platform classifier into a policy document. My MSc dissertation at Portsmouth (Distinction) built a cross-platform framework over 279,000 posts on four platforms and found the platform-divergence problem empirically; the methods paper with Dr Haig turned that into measurement theory. I run drift monitoring and provenance logging in production. What this project adds is the extremism construct, the Norwegian institutional setting, and a practitioner partner with a live need.

What I do not bring. I have no background in extremism or terrorism studies; I have read the literature above closely and will need the Department's and NIVE's expertise on the construct itself, which is why the codebook is co-developed rather than imported. I do not speak Norwegian. I have turned that into a research question in Study 1, where the cross-language invariance test asks whether the Norwegian and English instruments measure the same thing at all, but the interviews and the NIVE collaboration will need the Department's help and my own study of the language, which I will begin on arrival. I have not conducted qualitative interviews in research, and Study 3 is sized and supervised accordingly. My daily working language is Python; my R is functional, and I am working through the invariance and linking models from our paper in lavaan so that the analysis in Study 1 is done in the tools the field uses.

I am applying to this position rather than to a computer-science post because the question is not how to build a better classifier. It is what a number produced by a classifier means once it enters a system of platforms, agencies and policies that each read it differently, and how to make that number honest enough to act on. That is an Information Systems question, and Kristiansand, with NIVE across the road, is the place to answer it.

10. EXPECTED CONTRIBUTIONS

To Information Systems. A sociotechnical account of how the numbers that describe online extremism are produced, in which the classifier is treated as an affordance-bearing artefact whose outputs are actualised differently under platform, regulatory, practitioner and scientific logics. This extends the agenda set out by Risius and colleagues with a mechanism for one of its central puzzles: why estimates of the same phenomenon disagree across studies and across institutions, and why they travel so easily despite disagreeing.

To method. The first fitted invariance and generalizability analyses of extremism classifiers with platform, language and annotator standpoint as grouping variables and facets; an estimator for amplification that carries misclassification and drift through to the interval; and a shadow-anchored bridging design that lets a monitoring series survive vendor model updates. Each is a reusable procedure rather than a one-off result, and each will be released with code.

To practice. A validity protocol and reporting standard for extremism monitoring, developed with NIVE and evaluated in use, with direct relevance to the centre's mandate to develop a common risk assessment tool. The protocol specifies what a prevalence or risk figure must carry with it to be interpretable, which is the practical form of the thesis argument.

To the Norwegian evidence base. A Norwegian-language extremism corpus and instrument-exchangeability table built with properly resourced Norwegian models, and the first cross-language invariance test between Norwegian and English instruments in this domain, which answers a question every bilingual monitoring effort in the Nordic region currently assumes away.

11. SUPERVISION, COLLABORATION AND DISSEMINATION

The project is designed for supervision within the Department of Information Systems, with the affordance and institutional-logics strand drawing on the Department's established work in that tradition, and it asks for a co-supervisor for the interpretive study in Study 3. The measurement strand is where I am most self-sufficient; the extremism construct and the Norwegian institutional setting are where I most need the Department and NIVE. I would propose a standing working relationship with NIVE from the first months, structured around the codebook workshops in Study 1, the document and interview work in Study 3, and the protocol build in Study 4, with the centre's classification and clearance rules binding throughout. Dr Ella Haig at Portsmouth, co-author of the methods paper, has agreed to continue as an external collaborator on the measurement strand. The four papers map onto the four studies. Code, codebooks, the instrument-exchangeability table and the protocol will be released under open licences on publication, and aggregate findings will be written up for NIVE and for the Ministry of Justice and Public Security's follow-up to the Extremism Commission in a form practitioners can use. I would welcome the four-year track with 25 percent career-promoting work, and I would expect to contribute to teaching in data engineering, platform data collection and measurement for social science, which are the parts of the Department's curriculum my background serves best.

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